GEOG41160 – Assignment 2

Spatial Autocorrelation and Hotspots

1. Overview

In this task you will write a Python workflow that analyses where people in Dublin City extend their buildings based on the density of planning applications for extensions by Electoral Division (ED), perform global spatial autocorrelation, and analyse local clusters and hotspots. You will also practise working on one shared Python codebase, with each person's contribution visible in the Git history.

2. Data

Download both datasets, as GeoPackages, from the original sources.

- Planning applications: the Department of Housing, Local Government and Heritage's *Planning Application Points*, the national database of planning applications from all 31 councils (data.gov.ie, CC BY 4.0), at https://data-housinggovie.opendata.arcgis.com/datasets/housinggovie::planning-application-points. Save it as `data/planning/planning_applications_national.gpkg`.
- Electoral Divisions: Tailte Éireann's *CSO Electoral Divisions – National Statistical Boundaries – 2022 – Ungeneralised*, all 3,420 EDs (data.gov.ie, CC BY 4.0), at https://data-osi.opendata.arcgis.com/datasets/osi::cso-electoral-divisions-national-statistical-boundaries-2022-ungeneralised. Save it as `data/boundaries/eds_national.gpkg`.

3. Questions

There are three questions, and for each response, write each question as its own script inside folder `scripts/`: `Q1_ExtensionDensity.py`, `Q2_GlobalMoran.py`, and `Q3_LISA_GiStar.py`. There is a sample template on Brightspace (`geog41160_assignment2_script_template.py`) that you are free to use. Every script should:

- Start with a *docstring* (text between triple quotes, `"""`) holding your written answers to the question, numbered to match its items, quoting the values the script prints;
- Use relative paths (for example `data/planning/planning_applications_national.gpkg`), so it runs from a *fresh copy* of the repository. Use `Path` to define relative paths;
- For each step, add additional comments with short `#` comments.

Question 1. Extension density by ED (5 marks)

1. Create a DuckDB spatial database file, `outputs/extensions.duckdb`. Using SQL, import the two tables from the national files into the database, and report the number of rows remaining after each filtering condition:
   - `eds`: the EDs of Dublin City (`COUNTY_ENGLISH = 'DUBLIN CITY'`);
   - `extensions`: the planning applications received by Dublin City Council in the complete calendar years 2019–2025, where `DevelopmentDescription` contains the word "extension", regardless of capitalisation.
2. Print the CRS, number of features, and geometry type of both layers. Reproject the applications to the EDs' coordinate system (EPSG:2157) in SQL and explain why the analysis needs Irish Transverse Mercator rather than Web Mercator.
3. Spatially join the applications to the EDs in SQL, and count the applications in each ED. Report how many applications fall outside every ED, and why.
4. Calculate the density of applications in each ED (applications per km^2), and save the result as `DublinED_ExtensionDensity.gpkg`.
5. Map the density, and save the map as `ExtensionDensity_DublinED.png`.

Question 2. Global spatial autocorrelation (5 marks)

1. Check the distribution of extension density (histogram and skewness). If it is strongly skewed, transform it, and justify your choice.

2. Build spatial weights, and state which definition of neighbour you used, and why.
3. Compute Global Moran's I, with a fixed random seed and a stated number of permutations, for the raw density and for your chosen variable. Report I and p for both.
4. Interpret the results and state if the pattern clustered, dispersed, or random, and does the transformation change the conclusion? Why?

Question 3. Local clusters and hotspots (8 marks)

1. Compute Local Moran's I (LISA) with at least 9,999 permutations and a fixed seed. Classify EDs as High-High, Low-High, Low-Low, High-Low, or Not Significant at $p \leq 0.05$.
2. Apply a false discovery rate correction (`esda.fdr`), and report how many EDs remain significant.
3. Map the LISA classes with a legend, and save the map as `LISA_ExtensionDensity_DublinED.png`.
4. Compute Getis-Ord Gi*, classify hotspots ($Z > 1.96$) and coldspots ($Z < -1.96$), apply the false discovery rate correction to two-sided p-values.
5. Map the hotspots and coldspots with a diverging colour scheme, and save the map as `GiStar_ExtensionDensity_DublinED.png`.
6. Compare Gi* with LISA. Do they agree or and why?

4. Teamwork and Git (5 marks)

Decide on the roles in Sprint 0. Each lead role stays with one person throughout (in a team of 4, two people share one); the coordinator role rotates every sprint, so everyone coordinates at least once.

Role	Leads
Coordinator (rotates)	each sprint's plan, the README and sprint log; checks the scripts run from a fresh copy
Data and CRS lead	Question 1, and the shared, cleaned data
Spatial statistics lead	Question 2, and LISA in Question 3
Mapping and hotspot lead	all maps, and Gi* in Question 3

Work in four short sprints starting with Sprint 0 for setup, followed by one sprint for each question.

Set-up (Sprint 0).

- Create a free GitHub organisation, `UCD-GEOG41160-2026-Team-N` (your team number), with every member as an Owner, and in it a private repository, `dublin-extensions-spatial-analysis`.
- Create the folders `data/`, `scripts/`, and `outputs/`, and a `README.md`. Each member makes at least one commit.
- Write a README that explains the project, the folders, the CRS, the order to run the scripts, where the data comes from (the sources, and the date you downloaded the planning file), the licences, and a short sprint log (date, what was done, what is next). Do not list roles in the README, they will be included in the reflection.

Marking: 1.0 repository structure, 1.0 README and sprint log, 3.0 commit history: commits in every sprint, clear messages, and every member's contribution visible.

5. Individual Reflection (7 marks)

Lastly, each team member should write 200–300 words on their role, and the sprints they coordinated; their main technical contribution, including one result from the part they led, one decision they made, and why; and one thing they learned from a teammate.

6. Submission

Agree one final version, at the last commit. In the end, every student submit the following files to Brightspace:

- the team files:
  - `TeamRepositoryLink.txt`, with the repository's address;
  - the three scripts, in `GEOG41160_Assignment2_TeamNumber.zip`;

  - `DublinED_ExtensionDensity.gpkg` and the three maps: `ExtensionDensity_DublinED.png`, `LISA_ExtensionDensity_DublinED.png`, `GiStar_ExtensionDensity_DublinED.png`;
  - any other CSV files or figures the scripts produce;
- their own reflection, `Reflection_StudentNumber.pdf` (on Brightspace only, never in the repository).

Keep the repository private until the deadline, then make it public within 24 hours after submission. It will be marked from the public page. Check that the repository contains no student numbers or other personal details.

7. Marks and Rules

	Question 1	Question 2	Question 3	Git	Reflection	**Total**
Marks	5	5	8	5	7	**30**

- The questions and Git are team marks, while the reflection is individual.
- Some teams may be asked follow-up questions after submission. Therefore you should be prepared to explain the code, maps, and results of the part you contributed.
- Generative AI tools may not be used for the code, the written answers, or the reflection. Their use is a breach of academic integrity.